Sl. No.	Topic	Concepts Covered	# of Hours
1	Java Fundamentals	Java platform and JDK, program structure, variables and constants, primitive and non-primitive data types, operators, user input, type conversion and casting	3
2	Control Flow	Conditional statements (if, if-else, nested if, switch), looping constructs (for, while, do-while), loop control statements (break, continue), pattern-based programs	3
3	Methods	Method creation and invocation, parameters and arguments, return values, method overloading, variable scope, code reusability through methods	3
4	Arrays and Strings	Single-dimensional and multi-dimensional arrays, array traversal and manipulation, String class methods, StringBuilder, common string-processing techniques	3
5	Object-Oriented Programming	Classes and objects, constructors, encapsulation, inheritance, polymorphism, abstraction, interfaces, object-oriented design principles	3
6	Exception Handling and File Handling	Runtime and checked exceptions, try-catch-finally, throw and throws, custom exceptions, reading from and writing to files	3
7	Collections Framework	List, Set and Map interfaces, ArrayList and HashMap implementations, collection traversal techniques, lambda expressions, Stream API basics (map, filter, forEach)	3
8	Git and GitHub	Repository creation and management, commits, branching, merging, resolving conflicts, remote repositories, pull requests and collaboration workflows	3
9	JDBC	JDBC architecture, drivers, database connections, PreparedStatement, ResultSet processing, implementing CRUD operations using Java	3
10	Spring Framework	Inversion of Control (IoC), Dependency Injection (DI), bean creation and lifecycle, Spring container, annotations-based configuration, Spring architecture overview	3
11	Spring Boot	Spring Boot project setup, project structure, layered architecture (Controller-Service-Repository), starter dependencies, application configuration, auto-configuration	3
12	REST APIs	REST principles, controllers, request mapping annotations, HTTP methods, request and response handling, JSON serialization/deserialization, API testing using Postman	3
13	Spring Data JPA and Hibernate	Entity mapping, repositories, relationships, CRUD operations, derived query methods, pagination and sorting, ORM concepts	3
14	Validation and API Documentation	DTOs, Bean Validation, centralized exception handling, custom error responses, API documentation using Swagger/OpenAPI	3
15	Spring Security and JWT	Authentication and authorization, Spring Security configuration, JWT authentication flow, security filters, role-based access control	3
16	Spring Boot CRUD Application	Building a complete CRUD backend, integrating Controller-Service-Repository layers, database integration, API design best practices, project organization	4
17	HTML	HTML document structure, semantic elements, hyperlinks, images, tables, forms and form controls, accessibility basics, browser developer tools	3
18	CSS	CSS selectors, box model, typography, spacing, positioning, Flexbox, CSS Grid, responsive design using media queries	3
19	JavaScript	Variables and data types, functions, scope, arrays and objects, ES6 features (destructuring, spread/rest), DOM manipulation, event handling, Promises, async/await, Fetch API, JSON handling	3
20	Introduction to React	React ecosystem, Vite setup, JSX syntax, functional components, props, state with useState, event handling, conditional rendering, rendering lists	3
21	React Hooks and Components	useEffect, component lifecycle concepts, parent-child communication, lifting state up, component composition, reusable component design, managing side effects	3
22	React Forms and Routing	Controlled components, form handling, client-side validation, React Router, navigation, route parameters, protected routes	3
23	React State Management and API Integration	Context API, Axios, REST API integration, loading and error states, local storage, authentication flow, token persistence, environment variables, production build process	3